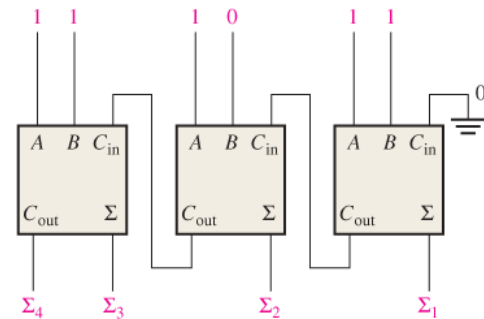


Problem Set 6

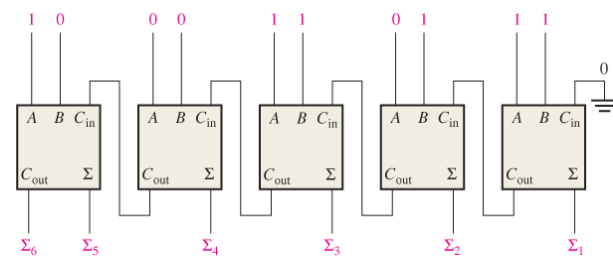
4. For the parallel adder in Figure 6–69, determine the complete sum by analysis of the logical operation of the circuit. Verify your result by longhand addition of the two input numbers.



	A3	A2	A1	
	1	1	1	
		1	1	1
+		1	0	1
	B3	B2	B1	

1	1	0	0
Σ_4	Σ_3	Σ_2	Σ_1

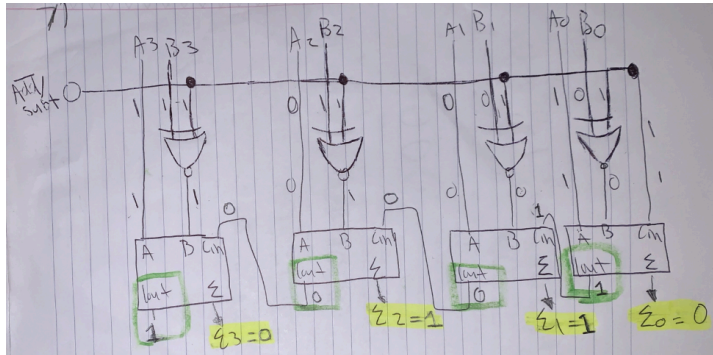
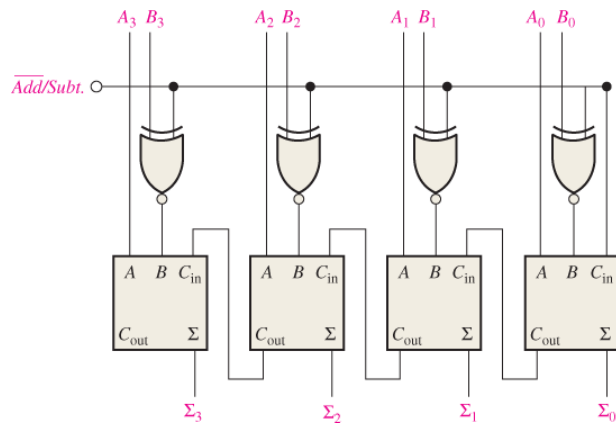
5. Repeat Problem 4 for the circuit and input conditions in Figure 6–70.



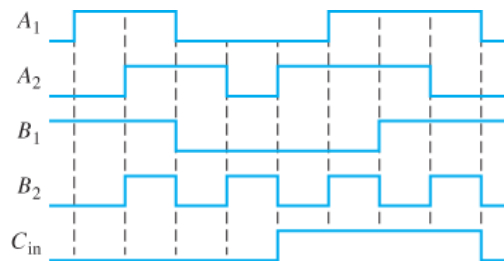
	A5	A4	A3	A2	A1	
		1	1	1		
	1	0	1	0	1	
+	0	0	1	1	1	
	B5	B4	B3	B2	B1	

0	1	1	1	0	0
Σ_6	Σ_5	Σ_4	Σ_3	Σ_2	Σ_1

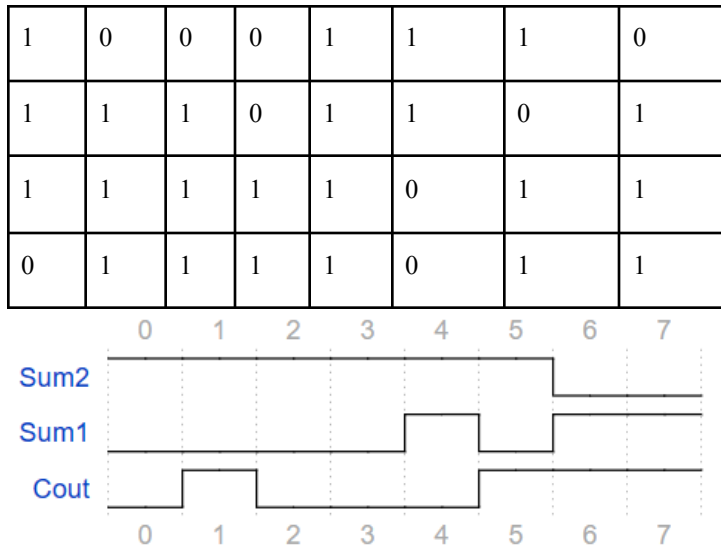
7. For the circuit in Figure 6-71, assume the inputs are $\overline{Add/Subt.} = 1$, $A = 1001$ and $B = 1100$. What is the output?



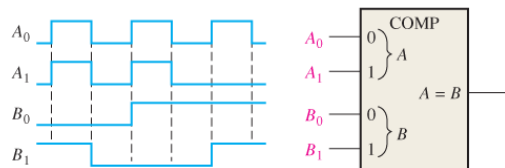
8. The input waveforms in Figure 6-72 are applied to a 2-bit adder. Determine the waveforms for the sum and the output carry in relation to the inputs by constructing a timing diagram.



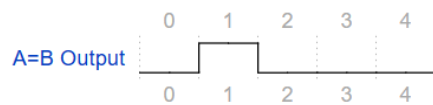
A2	A1	B2	B1	Cin	Sum 2	Sum 1	Cout
0	1	0	1	0	1	0	0
1	1	1	1	0	1	0	1
1	0	0	0	0	1	0	0
0	0	1	0	0	1	0	0



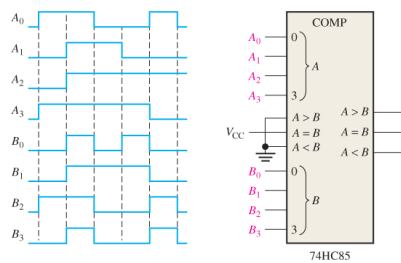
13. The waveforms in Figure 6–73 are applied to the comparator as shown. Determine the output (A=B) waveform.



0100 Only high when A=B



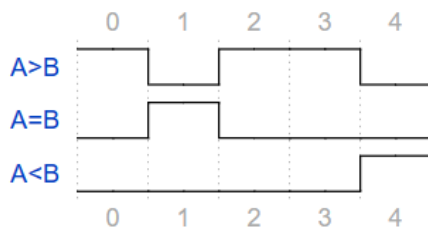
14. For the 4-bit comparator in Figure 6–74, plot each output waveform for the inputs shown. The outputs are active-HIGH.



A>B 10110

A=B 01000

A<B 00001



19. You wish to detect only the presence of the codes 1010, 1100, 0001, and 1011. An active-HIGH output is required to indicate their presence. Develop the minimum decoding logic with a single output that will indicate when any one of these codes is on the inputs. For any other code, the output must be LOW.

$$1010 = A \bar{B} \bar{C} \bar{D}$$

$$1100 = A \bar{B} \bar{C} \bar{D}$$

$$0001 = \bar{A} \bar{B} \bar{C} D$$

$$1011 = A \bar{B} \bar{C} D$$

$$X = (A \bar{B} \bar{C} \bar{D}) + (A \bar{B} \bar{C} \bar{D}) + (\bar{A} \bar{B} \bar{C} D) + (A \bar{B} \bar{C} D)$$

Simplified:

	00	01	11	10
00		1		
01				
11	1			
10			1	1

$$X = (A \bar{B} \bar{C} \bar{D}) + (\bar{A} \bar{B} \bar{C} D) + A \bar{B} \bar{C}$$

